SONY

Ver.1.0

IMX571BLR-J

Diagonal 28.3 mm (Type 1.8) CMOS Image Sensor with Square Pixel for Monochrome Cameras

Description

The IMX571BLR-J is a diagonal 28.3 mm (Type 1.8) CMOS active pixel type image sensor with a square pixel array and 26.11 M effective pixels. This sensor incorporates maximum 36 dB PGA circuit and 16-bit A/D converter. 16-bit digital output makes it possible to readout the signals of 26.11 M effective pixels at high-speed of 6.84 frame/s in still picture mode. In addition, it realizes 12-bit digital output for high-speed 4K moving picture by window readout mode. (Applications: Surveillance cameras, FA cameras, Industrial cameras).

Features

- APS size CMOS active pixel type dots
- Input clock frequency 72 MHz
- All-pixel readout mode (+ digital overlap drive)
  Various readout modes (*)
- Rolling shutter function moving picture mode
- H driver, V driver and serial communication circuit on chip
- +36 dB gain in CDS/PGA on chip (when A/D 16-bit, 14-bit, 12-bit)
- Built-in 11-bit/12-bit/14-bit/16-bit A/D converter
- 8 Lane SLVS-EC output
- Back-illuminated type

* Please refer to the datasheet for binning/subsampling details of readout modes.

STARVIS

* STARVIS is a registered trademark or trademark of Sony Group Corporation or its affiliates. The STARVIS is back-illuminated pixel technology used in CMOS image sensors for security camera applications. It features a sensitivity of 2000 mV or more per 1 μm^2 (color product, when imaging with a 706 cd/m^2 light source, F5.6 in 1 s accumulation equivalent), and realizes high picture quality in the visible-light and near infrared light regions.

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Device Structure

◆ CMOS image sensor	
◆ Image size	Diagonal 28.3 mm (Type 1.8)
◆ Total number of pixels	6280 (H) × 4264 (V) approx. 26.78 M pixels
◆ Number of effective pixels	6252 (H) × 4176 (V) approx. 26.11 M pixels
◆ Number of active pixels	6244 (H) × 4168 (V) approx. 26.02 M pixels
◆ Chip size	27.780 mm (H) × 22.302 mm (V)
◆ Unit cell size	3.76 µm (H) × 3.76 µm (V)
◆ Optical black	Horizontal (H) direction: Front 16 pixels, Rear 0 pixel Vertical (V) direction: Front 22 pixels, Rear 0 pixel
◆ Package	184 pin LGA

Image Sensor Characteristics

(Tj = 60 °C)

Item		Value	Remarks
Sensitivity (F8)	Typ.	5060 LSB	1/30 s integration
Saturation signal	Min.	14168 LSB	Left for 0.4 s

Basic Drive Mode

Drive mode	Number of active pixels	Max frame rate [frame/s]	Word length [bit]
Readout mode 0	6244 (H) × 4168 (V) approx. 26.02 M pixels	48.42	10
Readout mode 1	6244 (H) × 4168 (V) approx. 26.02 M pixels	18.82	12
Readout mode 3	6244 (H) × 4168 (V) approx. 26.02 M pixels	24.15	10
Readout mode 5	3122 (H) × 2084 (V) approx. 6.51 M pixels	37.55	12
Readout mode 6	2080 (H) × 1388 (V) approx. 2.89 M pixels	111.84	12
Readout mode 7	2080 (H) × 1386 (V) approx. 2.88 M pixels	111.99	12
Readout mode 10	2080 (H) × 1388 (V) approx. 2.89 M pixels	55.52	12
Readout mode 11	2080 (H) × 1386 (V) approx. 2.88 M pixels	55.52	12
Readout mode 12	2080 (H) × 832 (V) approx. 1.73 M pixels	245.08	12
Readout mode 13	2080 (H) × 832 (V) approx. 1.73 M pixels	245.08	12
Readout mode 14	2080 (H) × 594 (V) approx. 1.24 M pixels	339.77	12
Readout mode 15	2080 (H) × 594 (V) approx. 1.24 M pixels	339.77	12
Readout mode 16	2080 (H) × 462 (V) approx. 0.96 M pixels	432.44	12
Readout mode 17	2080 (H) × 462 (V) approx. 0.96 M pixels	432.44	12
Readout mode 18	2080 (H) × 320 (V) approx. 0.67 M pixels	611.99	12
Readout mode 19	2080 (H) × 319 (V) approx. 0.66 M pixels	611.99	12
Readout mode 20	2080 (H) × 166 (V) approx. 0.35 M pixels	1113.31	12
Readout mode 21	2080 (H) × 164 (V) approx. 0.34 M pixels	1125.28	12